CS 506 - Introduction to Quantum Computing

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1 Finishing Proof of Deutsch–Jozsa Problem

1.1 Stating $|\psi_2\rangle$ from previous results

$$|\psi_2\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{\vec{x}\,\in\,\{0,1\}^n} (-1)^{f(\vec{x})}\,|\vec{x}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{1}$$

where $\vec{x} = x_1\,x_2 \cdots x_n$ is an n-bit string.

1.2 Applying Hadamard Transformation : Obtaining $|\psi_3\rangle$

Apply Hadamard gate $H^{\otimes n}$ to $|\psi_2\rangle$:

$$|\psi_3\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{\vec{x}\,\in\,\{0,1\}^n} (-1)^{f(\vec{x})}\,H^{\otimes n}|\vec{x}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{2}$$

Sub-problem : Compute $H^{\otimes n}|\vec{x}\rangle$

We need to figure out what $H^{\otimes n}|\vec{x}\rangle$ equals.

Single-qubit case. If $x \in \{0,1\}$, we previously found :

$$H|x\rangle \;=\; \frac{1}{\sqrt{2}} \sum_{z\,\in\,\{0,1\}} (-1)^{zx}\,|z\rangle \tag{3}$$

★ Remember This!

Key Identity — for a single qubit $x \in \{0,1\}$:

$$H|x\rangle \;=\; \frac{1}{\sqrt{2}} \sum_{z\,\in\,\{0,1\}} (-1)^{zx}\,|z\rangle$$

Apply this independently to each qubit x_i to expand $H^{\otimes n}|\vec{x}\rangle$ as a product of single-qubit terms.

1.3 Notation Summary

Symbol	Meaning
$\vec{x} = x_1 x_2 \cdots x_n$	n-bit binary string
$f : \{0,1\}^n \to \{0,1\}$	Oracle function (constant or balanced)
$H^{\otimes n}$	Tensor product of n Hadamard gates
$(-1)^{f(\vec{x})}$	Phase kick-back from oracle U_f
$\lvert 0\rangle$, $\lvert 1\rangle$	Standard computational basis states

1.4 Expanding $H^{\otimes n}|\vec{x}\rangle$ as a Product

Let $\vec{x} = x_1 x_2 \cdots x_n$ where each $x_i \in \{0,1\}$. Since $H^{\otimes n}$ acts independently on each qubit :

$$H^{\otimes n}|\vec{x}\rangle \;=\; H|x_1\rangle\, H|x_2\rangle \cdots H|x_n\rangle \tag{4}$$

Applying the single-qubit identity to each factor :

$$H^{\otimes n}|\vec{x}\rangle \;=\; \left(\frac{1}{\sqrt{2}} \sum_{z_1 \in \{0,1\}} (-1)^{z_1 x_1} |z_1\rangle\right) \left(\frac{1}{\sqrt{2}} \sum_{z_2 \in \{0,1\}} (-1)^{z_2 x_2} |z_2\rangle\right) \cdots \left(\frac{1}{\sqrt{2}} \sum_{z_n \in \{0,1\}} (-1)^{z_n x_n} |z_n\rangle\right) \tag{5}$$

Collecting the n prefactors of $\frac{1}{\sqrt{2}}$ and merging all sums :

$$H^{\otimes n}|\vec{x}\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{z_1 \in \{0,1\}} \sum_{z_2 \in \{0,1\}} \cdots \sum_{z_n \in \{0,1\}} (-1)^{z_1 x_1 + z_2 x_2 + \cdots + z_n x_n}\, |z_1\rangle |z_2\rangle \cdots |z_n\rangle \tag{6}$$

Compact form. Define $\vec{z} = z_1 z_2 \cdots z_n$ with each $z_i \in \{0,1\}$. The exponent $z_1 x_1 + \cdots + z_n x_n$ is the **bitwise dot product** $\vec{z} \cdot \vec{x}$, giving the compact result :

$$\boxed{H^{\otimes n}|\vec{x}\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{z}\cdot\vec{x}}\, |\vec{z}\rangle} \tag{7}$$

★ Remember This!

Compact Notation — define $\vec{z} = z_1 z_2 \cdots z_n$, each $z_i \in \{0,1\}$. Then :

$$H^{\otimes n}|\vec{x}\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{z}\cdot\vec{x}}\, |\vec{z}\rangle, \qquad \vec{z}\cdot\vec{x} \;=\; \sum_{i=1}^{n} z_i x_i$$

This is the key result used to simplify $|\psi_3\rangle$ in the next step.

1.5 Substituting Back into $|\psi_3\rangle$

Substituting the compact result into $|\psi_3\rangle$:

$$|\psi_3\rangle \;=\; \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} \left[\frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{z}\cdot\vec{x} \bmod 2}\, |\vec{z}\rangle\right] \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{8}$$

Note. Since $(-1)^{\vec{z}\cdot\vec{x}}$ only depends on the parity of $\vec{z}\cdot\vec{x}$, we may write the exponent as $\vec{z}\cdot\vec{x} \bmod 2$. The two expressions are equivalent.

Simplifying. Merging the two prefactors $\frac{1}{\sqrt{2^n}} \cdot \frac{1}{\sqrt{2^n}} = \frac{1}{2^n}$ and combining both phase factors :

$$|\psi_3\rangle \;=\; \frac{1}{2^n} \sum_{\vec{x}\in\{0,1\}^n} \sum_{\vec{z}\in\{0,1\}^n} (-1)^{f(\vec{x}) + \vec{z}\cdot\vec{x} \bmod 2} |\vec{z}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \tag{9}$$

Observation. By measuring $|\psi_3\rangle$, we should be able to **distinguish between balanced and constant** oracle f (or we can call state).

1.6 Probability of Observing All Zeroes

We ask: with what probability do we observe $\vec{z} = 0^n = 00\cdots0$?

Setting $|\vec{z}\rangle = |0^n\rangle$ and noting $(-1)^{\vec{0}\cdot\vec{x}} = 1$ for all $\vec{x}$:

$$|\psi_3\rangle\Big|_{\vec{z}=0^n} \;=\; \left(\frac{1}{2^n} \sum_{\vec{x}\in\{0,1\}^n} (-1)^{f(\vec{x})} \right) |0^n\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \tag{10}$$

★ Remember This!

The **probability of observing** $\vec{z} = 0^n$ is the square of the amplitude :

$$P(\vec{z} = 0^n) \;=\; \left| \frac{1}{2^n} \sum_{\vec{x}\in\{0,1\}^n} (-1)^{f(\vec{x})} \right|^2$$

This single measurement distinguishes constant from balanced f — see next section.

1.7 Two Cases: Constant vs. Balanced

We now evaluate P_{0^n} for each case.

Case 1: $f(\vec{x})$ is constant. The value of $f(\vec{x})$ is the same for all $\vec{x}$, equal to either 0 or 1.

- If $f(\vec{x}) = 0$: $P_{0^n} = \left[\frac{2^n}{2^n}\right]^2 = 1$
- If $f(\vec{x}) = 1$: $P_{0^n} = \left[\frac{-2^n}{2^n}\right]^2 = 1$

In both sub-cases the probability of observing $|0^n\rangle$ is **1**.

Case 2: $f(\vec{x})$ is balanced. Exactly half of all inputs give $f(\vec{x}) = 0$ (contributing $+1$) and the other half give $f(\vec{x}) = 1$ (contributing -1). These cancel exactly :

$$\sum_{\vec{x}\in\{0,1\}^n} (-1)^{f(\vec{x})} = 0 \quad\Longrightarrow\quad P_{0^n} = \left[\frac{0}{2^n}\right]^2 = 0$$

If f is balanced we will **never** observe $|0^n\rangle$; the probability is 0.

★ **Remember This!**

Deutsch–Jozsa Decision Rule

$$P_{0^n} = \begin{cases} 1 & \text{if } f \text{ is } \textbf{constant} \\ 0 & \text{if } f \text{ is } \textbf{balanced} \end{cases}$$

A single observation/measurement suffices to determine whether f is constant or balanced — an exponential speed-up over any classical algorithm.

Wave-function interpretation (Schrödinger picture).

- **Constant** f : all phase factors $(-1)^{f(\vec{x})}$ are equal, so the amplitudes *constructively reinforce* each other at $|0^n\rangle$.
- **Balanced** f : exactly half the phases are $+1$ and half are -1, leading to complete *destructive interference* at $|0^n\rangle$.

2 Simon's Problem

*(Also known as the **Hidden Subgroup Identification Problem**.)*

2.1 Motivation : Random Sampling

Before stating the problem, we introduce the idea of **random sampling** — picking elements of a space uniformly at random.

Example 1 : Uniform binary sampling. Randomly pick $x_1, x_2, \ldots, x_n$ where each $x_i \in \{0, 1\}$ with equal probability :

$$\Pr[x_i = 0] = \tfrac{1}{2}, \qquad \Pr[x_i = 1] = \tfrac{1}{2}$$

This corresponds to random sampling over all 2^n possible n-bit binary strings.

Example 2 : Gaussian (normal) distribution. Random sampling from a Gaussian / normal distribution is another common example, used when the sample space is continuous rather than discrete.

Example 3 : Random direction on a hypersphere. Randomly pick a direction (unit vector) on the surface of a unit-radius n-dimensional hypersphere. In 2D this reduces to picking a random angle θ — equivalently, a random vector on the unit circle relative to a fixed reference vector.

2.2 The Core Challenge

We want to randomly sample from a space S **without knowing** S **explicitly**. We cannot simply enumerate S because it may be far too large to generate. To sample from such a space we need to know at least one *property* of the space.

2.3 Simon's Problem vs. Deutsch–Jozsa

Simon's problem is structurally similar to Deutsch–Jozsa :

- In Simon's problem we will **randomly sample a space without knowing the space**.
- The key difference from Deutsch–Jozsa is that the output of the oracle is **not just 1 bit but many bits**.

2.4 Formal Problem Statement

Suppose we have a **blackbox function** :

$$f : \{0,1\}^n \longrightarrow X \quad \text{(some set } X\text{, not just a single bit)}$$

We assume $X = \{0,1\}^n$ because f has 2^n possible inputs, so the output also needs 2^n values, which requires n bits to encode.

Given. There exists some string $\vec{s} \neq 0^n$ (if $\vec{s} = 0^n$ the solution is trivial) such that :

$$\forall\, \vec{x} \in \{0,1\}^n : \quad f(\vec{x}) \;=\; f(\vec{x} \oplus \vec{s})$$

where $\oplus$ denotes **bitwise XOR**. Explicitly, for $\vec{x} = x_1 \cdots x_n$ and $\vec{s} = s_1 \cdots s_n$:

$$\vec{x} \oplus \vec{s} \;=\; x_1 \oplus s_1,\; x_2 \oplus s_2,\; \ldots,\; x_n \oplus s_n$$

Goal. Find $\vec{s}$ using queries to f. We call $\vec{s}$ the **generator of the hidden space**.

★ Remember This!

Simon's Problem Setup

$$f(\vec{x}) = f(\vec{x} \oplus \vec{s}) \quad \forall\, \vec{x} \in \{0,1\}^n, \quad \vec{s} \neq 0^n$$

The goal is to find $\vec{s}$ — the hidden string — using as few queries to f as possible.

2.5 Classical vs. Quantum Query Complexity

Classical approach. To determine $\vec{s}$ correctly with probability $\geq \frac{2}{3}$, a classical algorithm needs $\Omega(2^{n/3})$ queries — exponential in n.

Quantum approach. Simon's quantum algorithm requires only $O(n)$ queries — an exponential speed-up.

2.6 Example : $n = 3$, $\vec{s} = 001$

Let $n = 3$, so both input and output are 3-bit strings, and suppose $\vec{s} = 001$. The condition $f(\vec{x}) = f(\vec{x} \oplus \vec{s})$ pairs up the $2^3 = 8$ inputs as follows :

$\vec{x}$	$\vec{x} \oplus \vec{s}$	f value
000	001	000
010	011	100
100	101	101
110	111	111

The pairs $\{000, 001\}$, $\{010, 011\}$, $\{100, 101\}$, $\{110, 111\}$ each share the same f value. In other words, the 2^n inputs are partitioned into **groups of 2** whose f values are equal :

$$\#\text{ of groups} \;=\; \tfrac{1}{2} \cdot 2^n \;=\; 2^{n-1}$$

2.7 Pre-calculations : Vector Space over $\mathbb{Z}_2^n$

Before moving to the Simon algorithm, we set up the mathematical structure we will need.

Consider the vector space of **n-bit binary strings** with addition modulo 2, denoted $\mathbb{Z}_2^n$. The coefficients are 0 or 1, and addition is bitwise XOR.

Example : $n = 3$. There are $2^3 = 8$ possible binary strings. The **basis vectors** are the strings with exactly one 1 :

$$\vec{e}_1 = 100, \quad \vec{e}_2 = 010, \quad \vec{e}_3 = 001$$

Every string can be written as a linear combination of these basis vectors with coefficients in $\{0, 1\}$ and addition mod 2. For example :

$$\begin{aligned} 111 &= 1(100) + 1(010) + 1(001) \pmod 2 \\ 000 &= 1(100) + 1(100) \pmod 2 \\ 011 &= 0(100) + 1(010) + 1(001) \pmod 2 \end{aligned}$$

2.8 Subspaces and the Hidden Subspace S

Consider 2 strings : $\vec{0}$ and $\vec{s}$. The set $S = \{\vec{0}, \vec{s}\}$ forms a **subspace** of $\mathbb{Z}_2^n$, with $\text{span}(S) = S$. No matter what linear combination you take of elements of S (with coefficients in $\{0, 1\}$ mod 2), you remain inside S :

$$\begin{aligned} 0 \cdot \vec{0} + 0 \cdot \vec{s} &= \vec{0} \\ 0 \cdot \vec{0} + 1 \cdot \vec{s} &= \vec{s} \\ 1 \cdot \vec{0} + 0 \cdot \vec{s} &= \vec{0} \\ 1 \cdot \vec{0} + 1 \cdot \vec{s} &= \vec{s} \end{aligned}$$

The basis vector of S is $\vec{s}$ alone, so $\dim(S) = 1$.

2.9 Orthogonal Complement $S^\perp$

The **orthogonal complement** of S in $\mathbb{Z}_2^n$ is defined as :

$$S^\perp = \left\{ \vec{z} \in \{0, 1\}^n \;\middle|\; \vec{s} \cdot \vec{z} \equiv 0 \pmod 2 \right\}$$

That is, $S^\perp$ consists of all n-bit strings $\vec{z}$ whose dot product with $\vec{s}$ is even. $S^\perp$ is orthogonal to $S = \{\vec{0}, \vec{s}\}$.

Dimension relationship. A fundamental result for $\mathbb{Z}_2^n$ gives :

$$\dim(S) + \dim(S^\perp) = \dim(\mathbb{Z}_2^n) = n$$

Since $\dim(S) = 1$, it follows that :

$$\dim(S^\perp) = n - 1$$

★ Remember This!

Key Dimension Result

$$\dim(S) + \dim(S^\perp) = n$$

For Simon's problem, $S = \{\vec{0}, \vec{s}\}$ has $\dim(S) = 1$, so the orthogonal complement satisfies $\dim(S^\perp) = n - 1$. The quantum algorithm will sample uniformly from $S^\perp$ to recover $\vec{s}$.